

ATOR EARTH

Global CPU, GPU, AI Mineral Extraction Master Inventory

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Abstract

This research log presents ATOR EARTH, an open inventory that traces extraction nodes feeding CPU, GPU and AI hardware. The work starts from a simple observation. The mineral chain behind chips is rarely shown as a continuous map. Tin solder from Bangka Belitung, nickel laterite and HPAL from Sulawesi and Halmahera, copper gold from Grasberg, cobalt from Katanga, high purity quartz from Spruce Pine, and rare earths from Bayan Obo are usually reported as separate commodities. ATOR EARTH brings them into one archival surface.

The inventory contains more than 100 nodes across Si, Li, Sn, REE, Co, Ni, Cu, Ta Nb, W, Ga, Ge, In, B, P, Sb, graphite, Mn, Al Ga, Ag and PGM families. Each node stores world region, country, site name, latitude longitude, mineral code, family, hardware relevance, extraction type, operator company, government authority, environmental risk, source url and source status. Methodology separates fact from inference. Origin is noted. Evidence is linked. Rival explanations are tested. Failure conditions are stated.

A large portion of operator_company is still marked Source needed. That mark is kept intentionally as a gap for beneficial ownership search under EITI and OECD Due Diligence Guidance Step 2. It is not an accusation. It indicates where corporate registry search via MODI Minerba IUP, AHU Online, OpenCorporates and SEC EDGAR 20F is still required, and where RMI RMAP smelter cross check is needed. All live data and maps are provided openly in this log for independent verification.

Keywords: critical minerals, nickel HPAL, tin Bangka, supply chain, beneficial ownership, OECD due diligence, EITI, environmental risk, open inventory

Live Access. All Data Provided

All data and live links are provided openly. No request for action is made. The reader can verify independently.

Research Log PDF v4:

https://deborahrammozes.github.io/ATOR.Institute./ator-earth/ATOR_EARTH_Research_Log_v4_Final.pdf

Map Dual Satellite plus Black archival:

https://deborahrammozes.github.io/ATOR.Institute./ator-earth/ator_earth_v2_dual.html

Map Offline archival never blank:

https://deborahrammozes.github.io/ATOR.Institute./ator-earth/ator_earth_v2_offline.html

Original Map v1: https://deborahrammozes.github.io/ATOR.Institute./ator-earth/ator_earth_global_cpu_gpu_mineral_map.html

CSV Inventory 100+ nodes: https://github.com/DeborahRamMozes/ATOR.Institute./blob/main/ator-earth/ator_earth_global_cpu_gpu_mineral_inventory.csv

GitHub Repository: <https://github.com/DeborahRamMozes/ATOR.Institute.>

Methodology

Sources are USGS Mineral Commodity Summaries, DOE Critical Minerals and Materials, Indonesia MEMR Mineral Yearbook 2023, and peer reviewed IJOG study on Bangka Belitung REE association. Each node carries claim status regional_node_needs_site_verification until field and registry verification. Export basis marks certainty: company_or_country_level versus country_level. Environmental risk is literature based, not field measurement. No fake certainty is allowed. The inventory follows evidence discipline with clear separation

between observed data and interpretation.

Indonesia Findings. Six Nodes

Indonesia is central to tin solder and nickel HPAL battery chain for AI hardware. Six nodes below are extracted from master CSV. Four of them still have Source needed in operator_company. That gap is the focus for corporate registry search.

node_id	Site	Lat Lon	Code	Operator Company Gap indicates BO search needed	Environmental Risk Literature based
sn_idn_bangka	Bangka Tin and REE Associated Mineral Region	-2.133, 106.117	Sn+REE	PT Timah plus artisanal mixed. Partial. Requires IUP and trader check	Offshore dredging, coastal turbidity, tailings, monazite REE association
sn_idn_belitung	Belitung Tin and REE Associated	-2.871, 107.953	Sn+REE	PT Timah plus artisanal mixed. Partial	Seabed disturbance, turbidity, land degradation
sn_idn_kundur	Kundur Tin Region	0.733, 103.433	Sn	PT Timah regional operators. Partial	Land and coastal disturbance
cu_idn_grasberg	Grasberg Papua Copper Gold	-4.053, 137.116	Cu+Au	Freeport Indonesia MIND ID. Company level	Tailings, mountain ecosystem impact, river water risk
ni_idn_sulawesi	Sulawesi Nickel Belt	-3.0, 121.5	Ni	SOURCE NEEDED. Gap for AHU and MODI check	Deforestation, laterite stripping, smelting pollution
ni_idn_obi_halmahera	Obi Halmahera Nickel Cobalt HPAL	-1.45, 127.6	Ni+Co	SOURCE NEEDED. Gap for AHU and MODI check	HPAL waste, acid leach tailings, deforestation

Shell Chain Extension

Current inventory stops at operator_company. v2 adds six steps. Mine Site, License Holder via ESDM MODI IUP, Trader Aggregator, Smelter Refiner via RMI RMAP list, Manufacturer, Tech Company. Step 2 is where beneficial ownership search is needed. Red flag criteria: two or more layers between mine and smelter with no disclosed beneficial owner, offshore jurisdiction, and no RMI audit. This follows OECD Step 2 risk assessment.

Limitations

All nodes are regional until site level enrichment. Operator Source needed means unverified, not proven shell. Environmental risk is literature based, not field measurement. Coordinates are approximate, rounded for safety on high risk sites. Corporate registry data changes over time. This log does not name private individuals. It provides gap analysis for further verification under EITI.

Citation

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